

Anurag Pradhan

anuragpradhancb@gmail.com | +91-7207068063

[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

SUMMARY

Prefinal-year Computer Science student at VIT Chennai with hands-on experience in Deep Learning and Machine Learning. Skilled in designing, training, and deploying efficient CNNs, Transformers, LLMs and Vision-Language Models, with expertise in PyTorch, data preprocessing, optimization, and on-device inference. Published paper at IJCNLP–AAACL 2025, focusing on accessibility-driven evaluation of SmolVLM2-500M-Video-Instruct.

EDUCATION

- Vellore Institute of Technology** Chennai, India
Bachelor of Technology in Computer Science; CGPA: 7.83/10 *August 2023 – June 2027*
- Sri Venkateswara Jr. College** Visakhapatnam, India
Class 12th (MPC); Score: 88.4/100 *July 2021 – June 2023*
- Sri Chaitanya Techno School** Visakhapatnam, India
Class 10th Score: 93.6/100 *February 2020 – March 2021*

SKILLS SUMMARY

- Programming Languages:** Python, C, C++, Java, Kotlin
- Core Expertise:** Vision-Language Models (VLMs), Large/Small Language Models, EdgeAI & On-Device Inference, Agentic AI, Generative AI, Computer Vision, Transformers
- Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, llama.cpp, SmolVLM2, BEiT, BLIP, OpenCV, LangChain
- Tools & Platforms:** Git, Docker, CUDA, ONNX, Weights & Biases, Hugging Face Hub, Android/Compose (JNI), ARM NEON/SVE
- Research Areas:** LLM Weight Quantization, VLM Accessibility Evaluation, Action Recognition, EdgeAI
- Soft Skills:** Leadership, Technical Writing, Event Management, Time Management

EXPERIENCE

- National Aluminium Company Ltd (NALCO)** Damanjodi, Odisha
Computer Science Intern *June 2025 – July 2025*
 - Contribution:** Worked on RespiGuard AI, an AI-based respiratory safety system for mining environments. Improved hazard detection reliability using deep-learning-based visual analysis.
 - Tech Stack:** Custom YOLOv5, PyTorch, Data Augmentation, Model Evaluation
- Team Ignition, VIT Chennai** On Site
Flight Software Lead *Apr 2024 – Apr 2026*
 - Contribution:** Developed flight software for model rockets, including fin actuator control, stability algorithms, onboard video stabilization, and Kalman-based state estimation.
 - Tech Stack:** Arduino, C/C++, PID Control, Image Processing, Deep Learning

PROJECTS

- DrishhtiAI — Android App:** Architected a high-performance Android app for offline multimodal Q&A using SmolVLM2 via llama.cpp. Engineered efficient JNI/C++ pipelines for low-latency RGB frame extraction and real-time visual processing. Implemented a sensor fusion engine combining YOLO-World and Depth-Anything for precise, object-specific distance estimation and spatial analysis.
Tech: VLMs, Quantization, Llama.cpp, GGUF, Kotlin
- CMCA-ActionNet – Spatio-Temporal Multimodal Action Detection:** Engineered a multimodal action detection system achieving 84% accuracy on JHMDB. Developed a **Cross-Modal Causal Attention (CMCA)** module to dynamically fuse RGB, optical flow, and MediaPipe landmarks. Leveraged bidirectional causal temporal attention and a two-branch uncertainty head for robust sequence reasoning and calibrated confidence estimation. Tech: PyTorch, Transformers, MediaPipe, Optical Flow, Causal Attention, Uncertainty Quantification
- Brain Tumor Segmentation:** Built a BEiT-transformer segmentation pipeline with skip connections and attention-driven decoding. Trained with Albumentations and CE–Dice hybrid loss for stable convergence. [\[GitHub Repo\]](#)
Tech: Python, PyTorch, Vision Transformers, BEiT, Semantic Segmentation
- JanaSathi: Odia E-Governance Chatbot:** Designed a bilingual RAG-based chatbot for Odisha government schemes using OpenRouter LLMs and MiniLM embeddings. Integrated Odia translation with Cohere Command-R+. Achieved 2nd place in AMD-sponsored OdiaGenAI Hackathon 2025. [\[GitHub Repo\]](#)
Tech: Python, Transformers, LangChain, Retrieval-Augmented Generation

CONFERENCE PUBLICATIONS

- Towards Blind and Low-Vision Accessibility of Lightweight VLMs and Custom LLM-Evals:** S. S. Baghel, Y. P. S. Rathore, A. Pradhan. Accepted at the 14th International Joint Conference on Natural Language Processing & the 4th Asian Chapter of ACL (IJCNLP–AAACL 2025), Mumbai, India. [\[Paper Link\]](#)